
Spectroscopic Characterization of Transmon-Resonator cQED Systems

Readout Power Calibration, Photon-Number
Splitting, and Anharmonicity Analysis

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Motivation :

- Precise characterization is essential for superconducting qubit operation.
- Spectroscopy enables extraction of qubit transition frequencies and nonlinear effects.
- Drive and readout powers strongly affect qubit response and measurement accuracy.
- Adaptive spectroscopy can improve transition detection and parameter extraction.

Problem Statement :

Analyze power-dependent spectroscopy and extract higher-level transition dynamics in transmon-resonator cQED systems.

Objectives of the Work:

- Perform frequency-domain spectroscopy of superconducting qubit systems
- Estimate the optimal readout power for spectroscopy measurements
- Study power-dependent effects using two-tone spectroscopy
- Analyze photon-number splitting and power broadening phenomena
- Estimate optimal drive power for qubit excitation
- Extract transmon anharmonicity using higher-level transitions

Literature Review:

Dispersive Readout & Resonator Spectroscopy

- A. Blais et al., Phys. Rev. A (2004) — Circuit QED dispersive framework
- B. Suri, PhD Thesis (2015) — Transmon coupled to lumped-element resonator
- Jaynes–Cummings model in dispersive regime for readout
- Photon-number splitting in strong dispersive regime

Qubit Spectroscopy

- J. Koch et al., Phys. Rev. A (2007) — Transmon spectroscopy
- Low-power spectroscopy for extracting natural linewidth (T_2^*)

Power Spectroscopy & Broadening

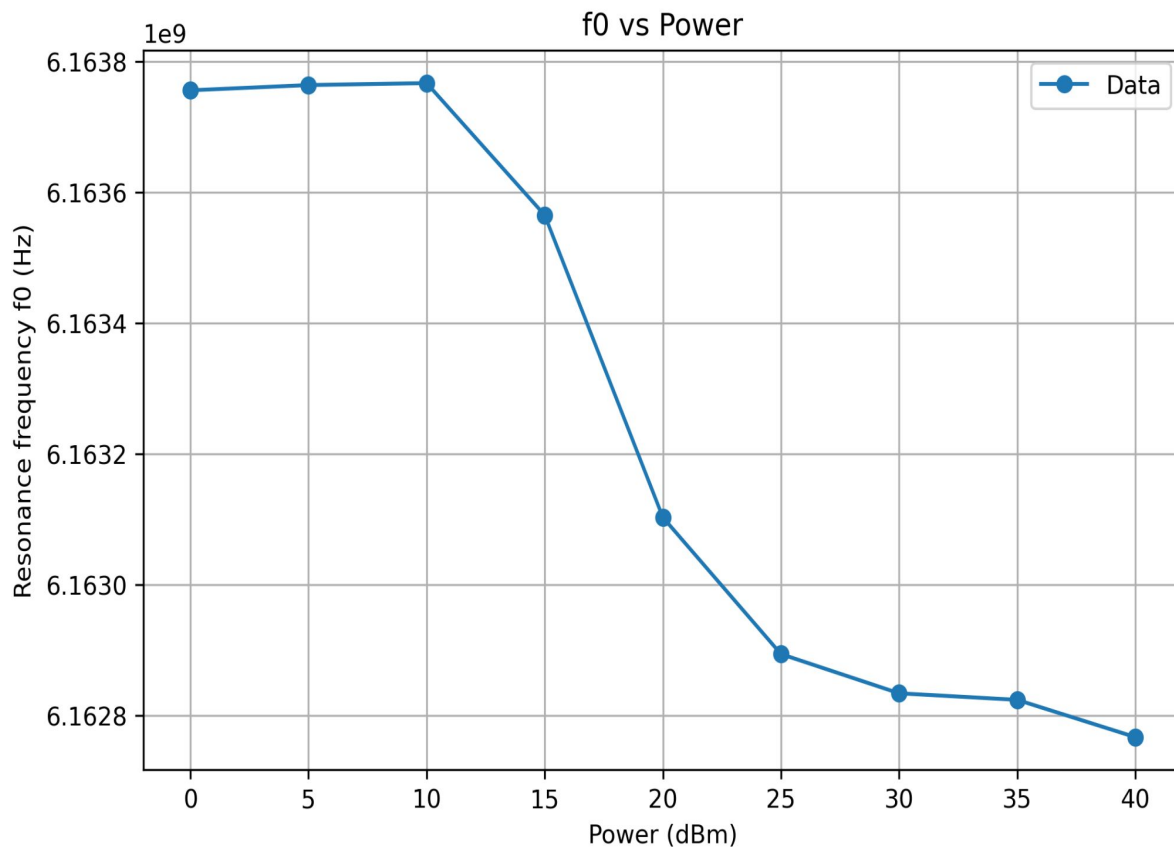
- M. Boissonneault et al., Phys. Rev. A (2009) — photon-induced dephasing
- Power-dependent linewidth broadening under strong drive

Theoretical Foundation

- Quantum Measurement course notes
- Quantum Optics course note

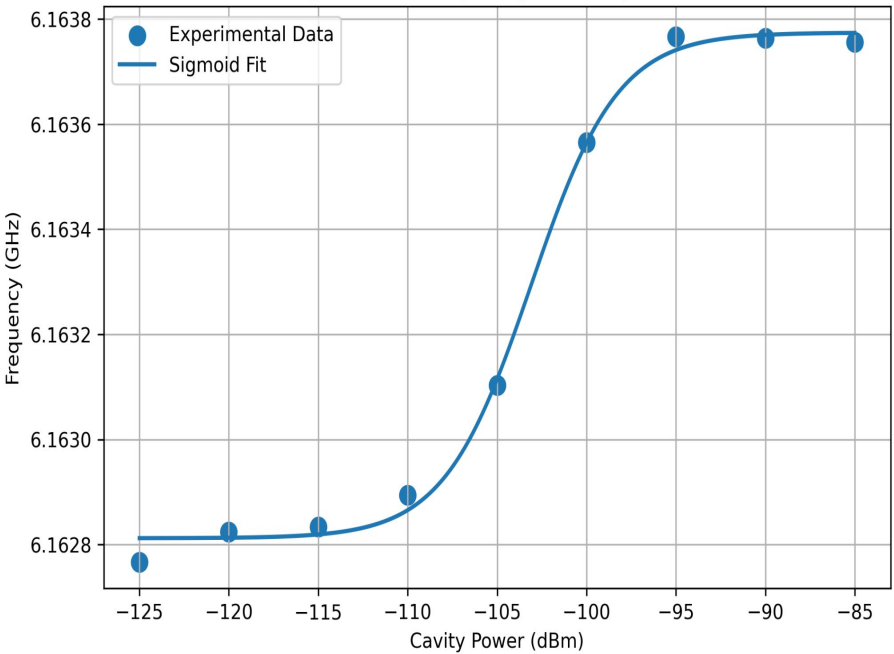
Optimal Readout Power:

- Resonator transmission was measured under different readout powers
- Increasing readout power modifies transition visibility and resonator response
- An optimal readout power region was identified for spectroscopy measurement

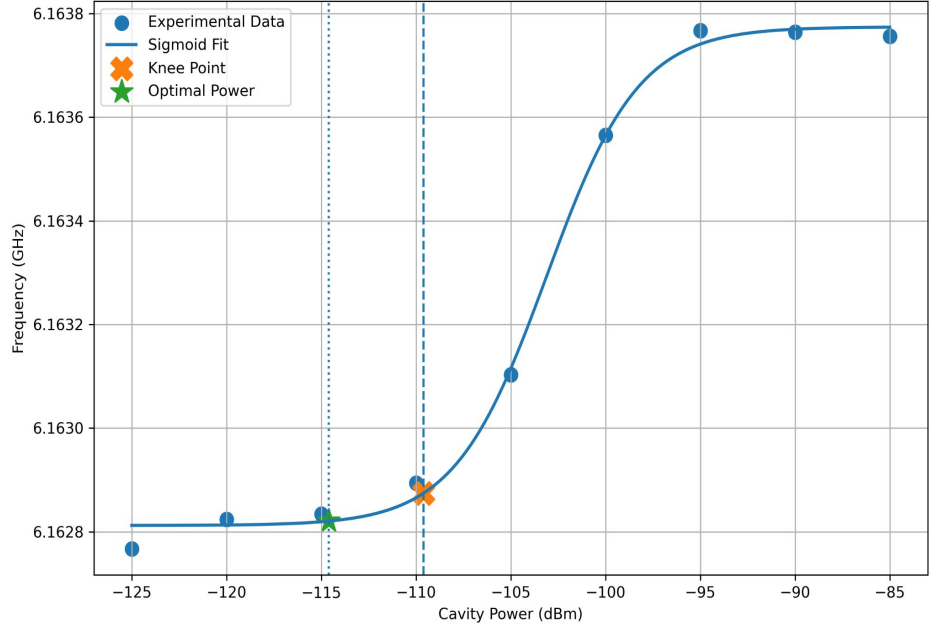


Sigmoid Fit and Optimal Power:

Sigmoid Fit: Frequency vs Cavity Power



Sigmoid + Knee Detection for Optimal Readout Power

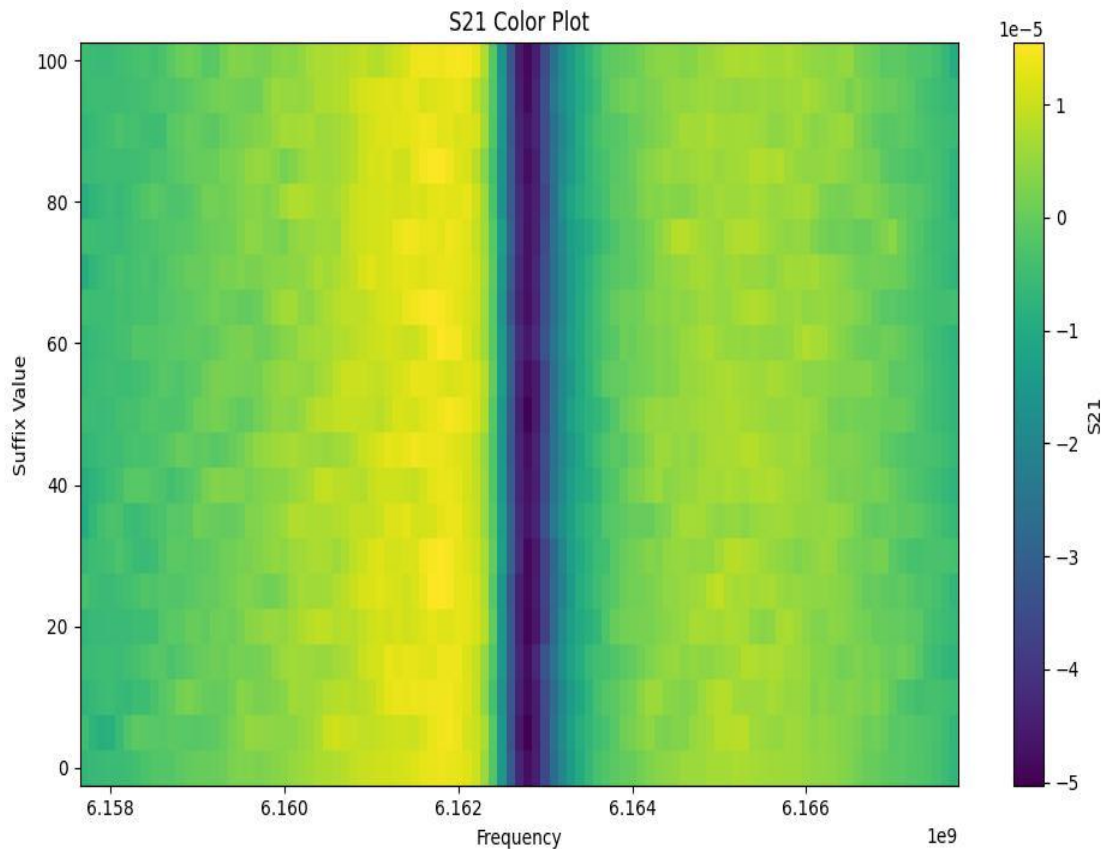


RESULTS:

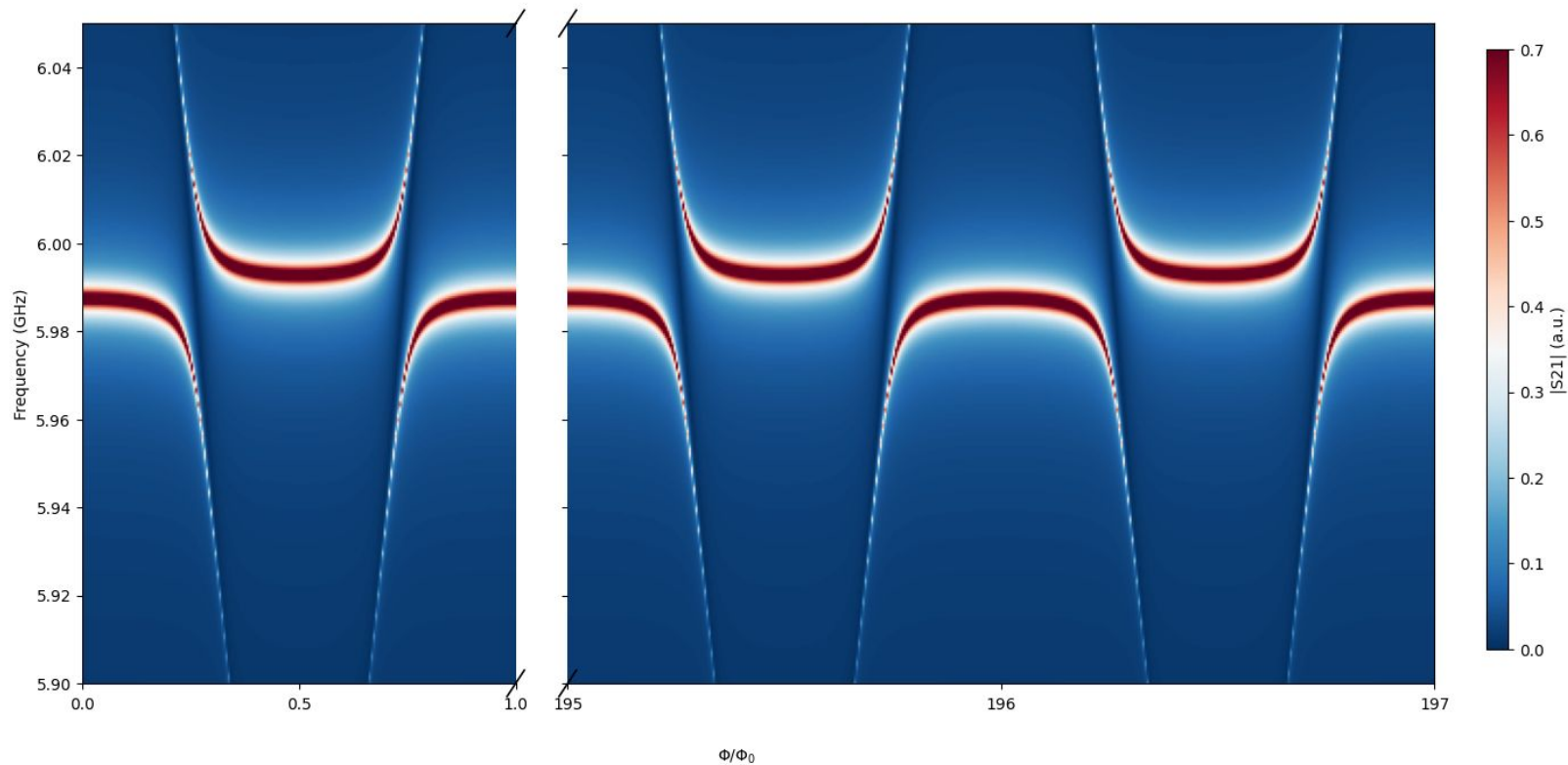
Knee Power = -109.61 dBm
Knee Frequency = 6.162874 GHz
Optimal Power = -114.61 dBm

Experimental Limitation:

- Readout power dependence was experimentally measured on a fluxonium device
- Experimental data shows a weak resonant feature near the expected frequency
- Clear qubit-resonator response could not be consistently resolved
- This indicates instability in the device response at higher drive powers
- This motivates a simulation-based analysis



Experimental readout response showing weak resonance feature



Simulated expected resonator response in the absence of experimental instability

Power Dependent Resonator Response(Dispersive Regime):

$$H_{JC} = \hbar\omega_r a^\dagger a + \frac{\hbar\omega_{ge}}{2}\sigma_z + \hbar g_{ge}(a\sigma^+ + a^\dagger\sigma^-)$$

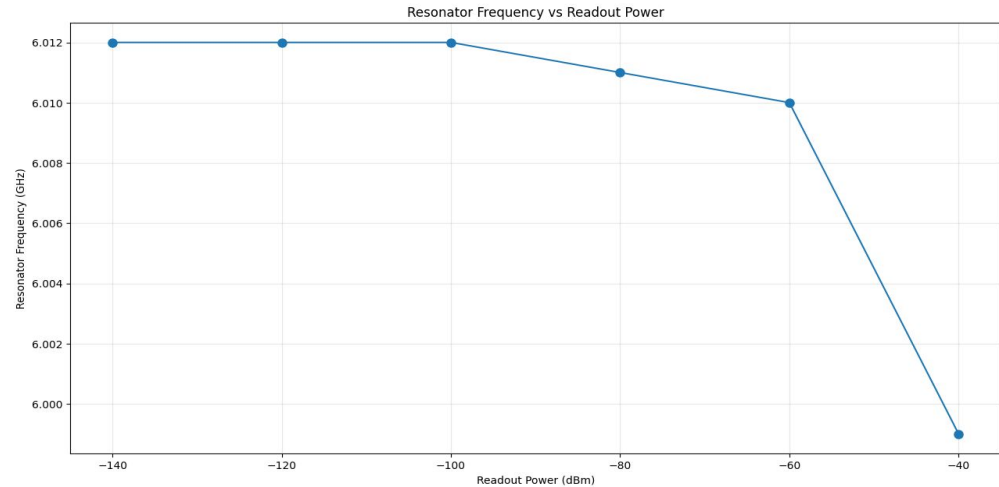
- System modeled in dispersive regime using circuit QED framework

- Bare resonator frequency: $\omega_c / 2\pi = 6$ GHz

- Dispersive coupling induces qubit-state dependent frequency shift

$$\omega_r(\text{shifted}) = \omega_r + \chi\langle\sigma_z\rangle$$

- $\chi = 100$ MHz

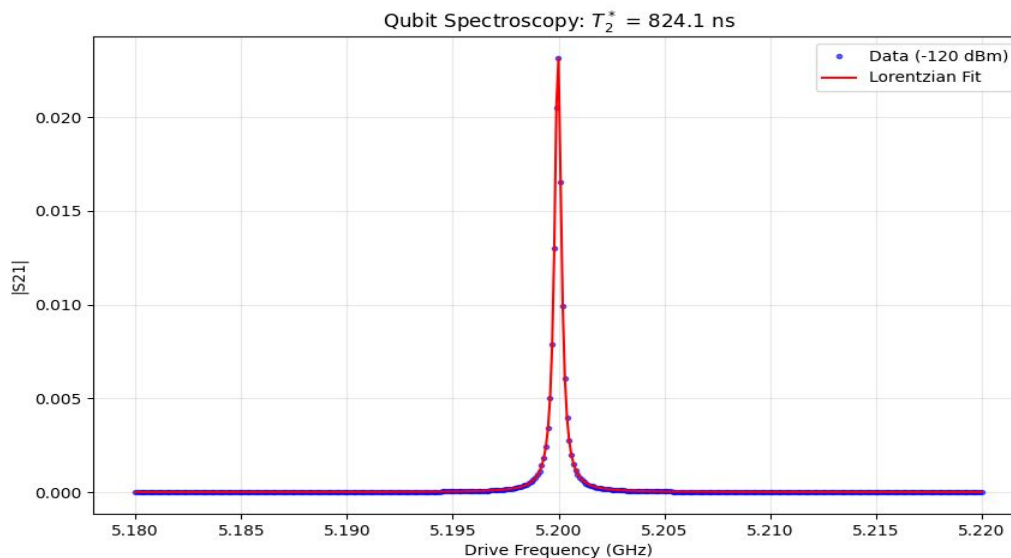


Readout parameters:
Readout frequency = 6.0125 GHz
Readout power = -100 dBm

Low Power Qubit Spectroscopy:

Measurement Conditions:

- Readout tone fixed at resonator frequency
- Readout power: -100 dBm
- Weak qubit drive applied
- Drive frequency swept across qubit transition range



Results:

Qubit transition frequency: 5.199974 GHz

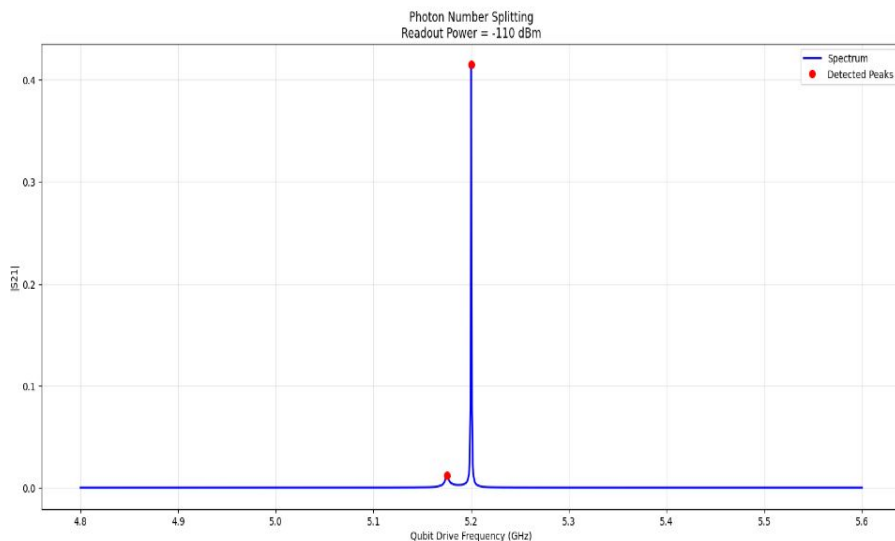
Intrinsic linewidth: 0.386 MHz

$T_2^* = 1 / (\pi \times \text{linewidth})$

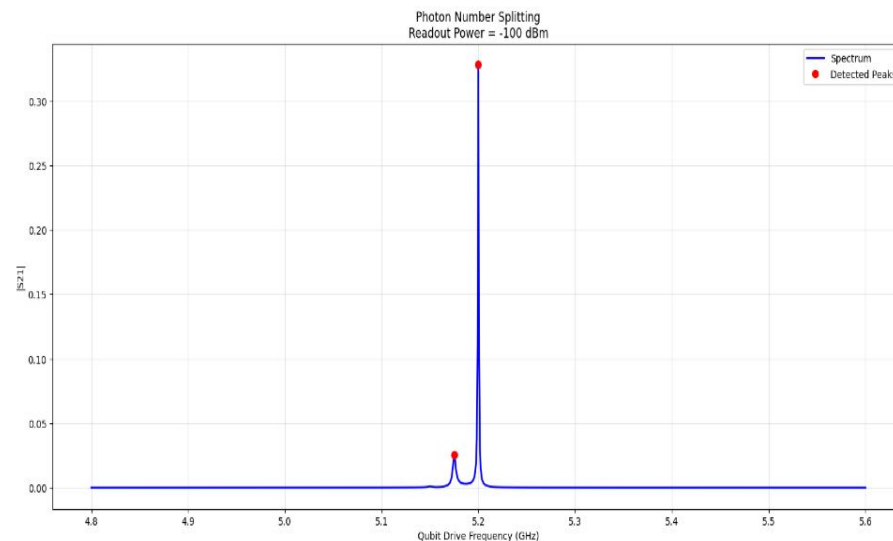
$T_2^* \approx 0.82 \mu\text{s}$

Observation of Photon-Number Splitting:

-110 dBm



-100 dBm



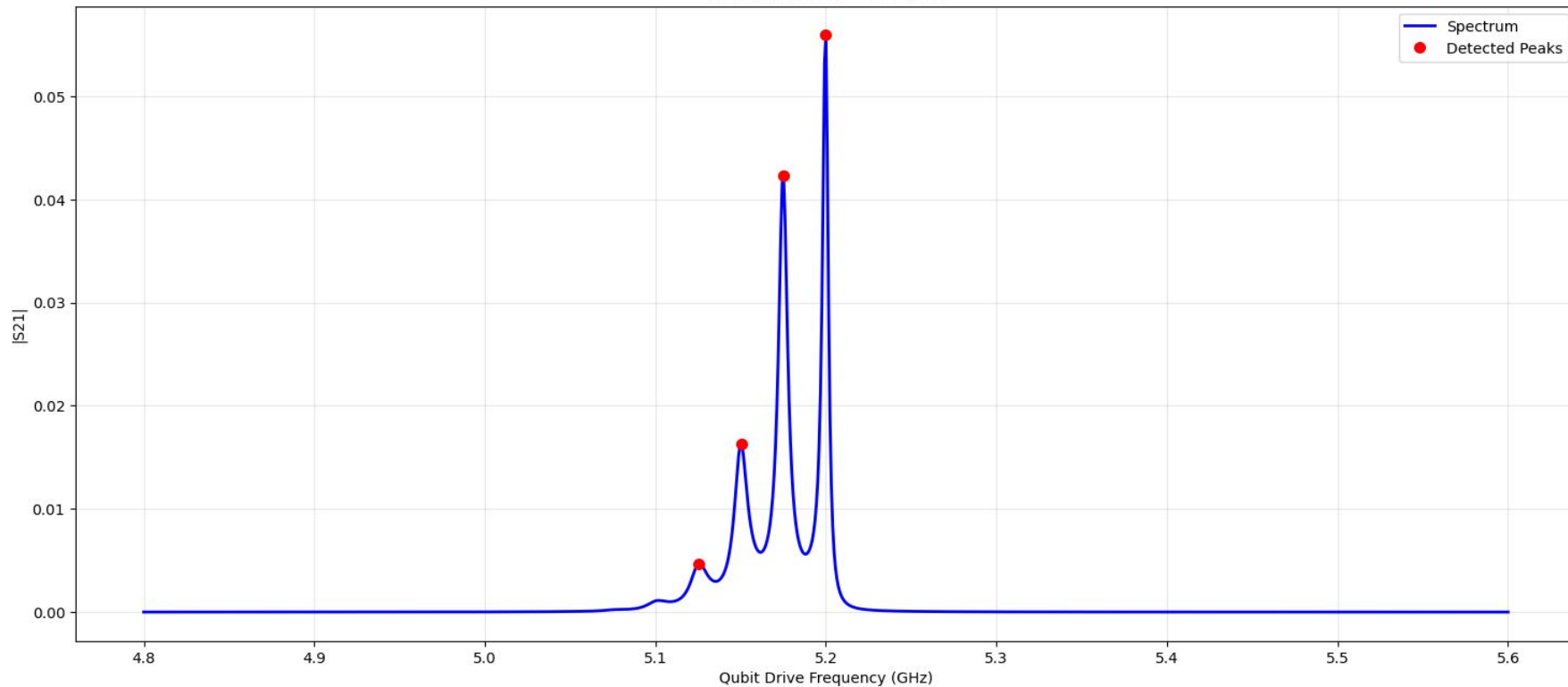
Measurement Conditions:

- Readout power increased gradually
- Readout frequency fixed near resonator frequency
- Weak qubit spectroscopy tone

$$\omega_q(n) = \omega_q + 2\chi n$$

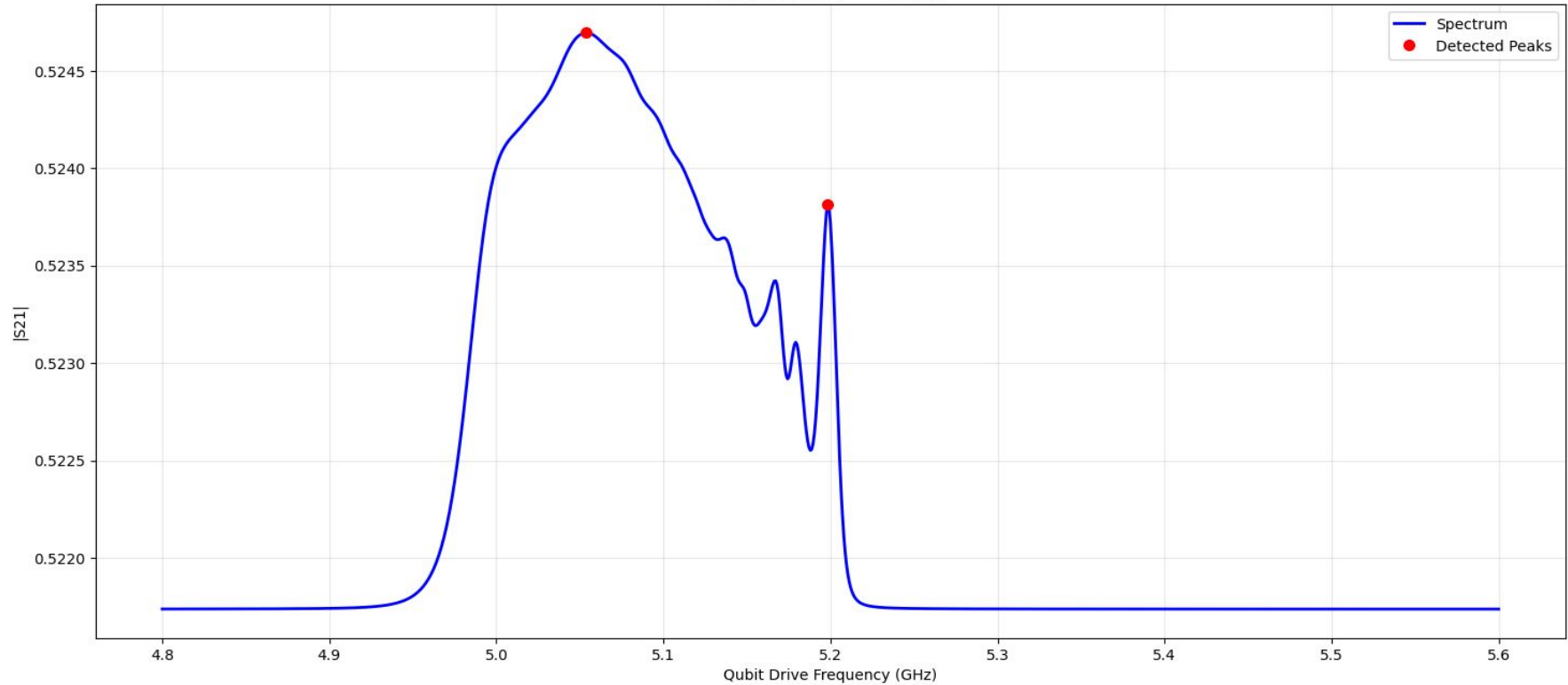
Expected peak spacing ≈ 25 MHz

Photon Number Splitting
Readout Power = -90 dBm



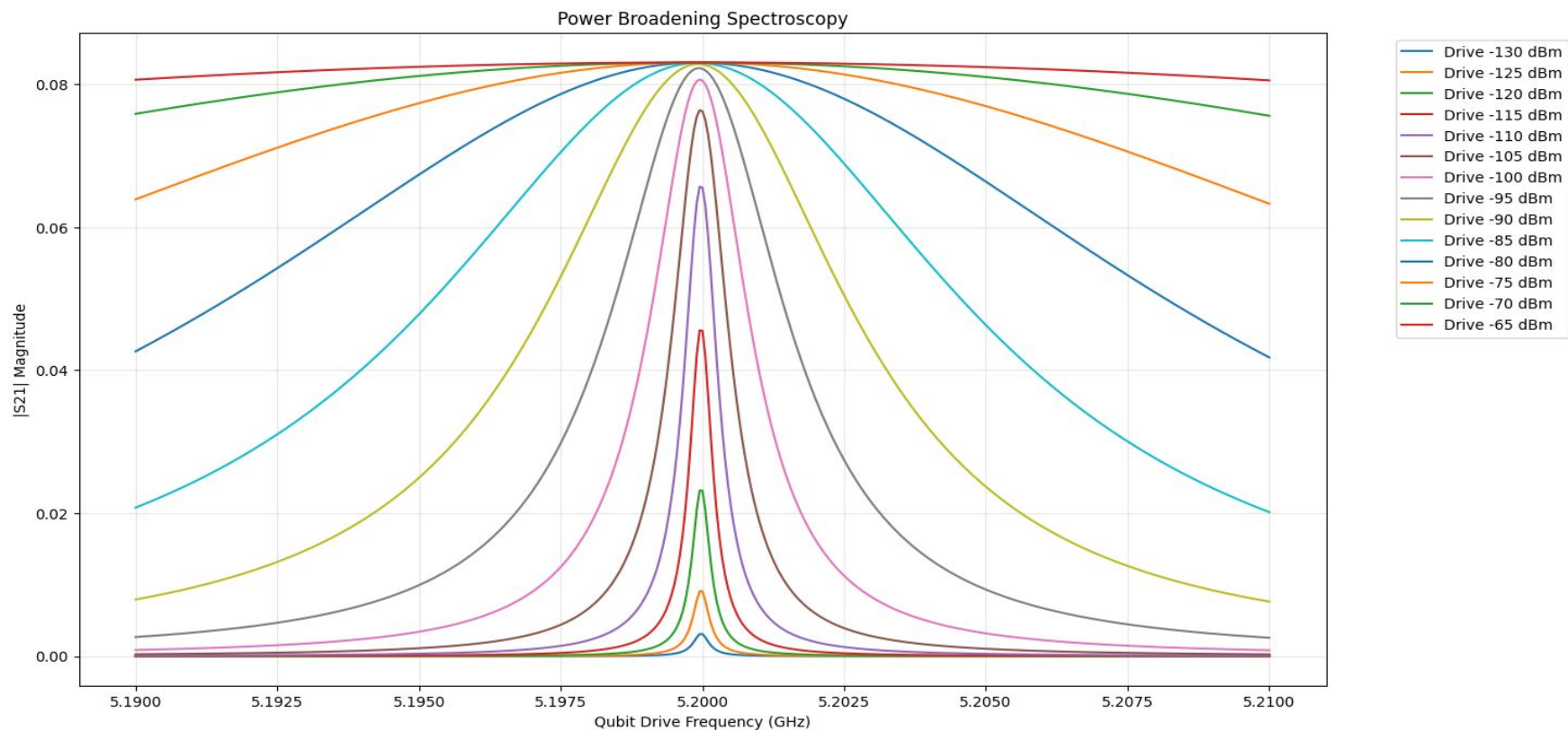
Emergence of photon-number splitting at approx
 $\bar{n} \approx 1$

Photon Number Splitting
Readout Power = -80 dBm

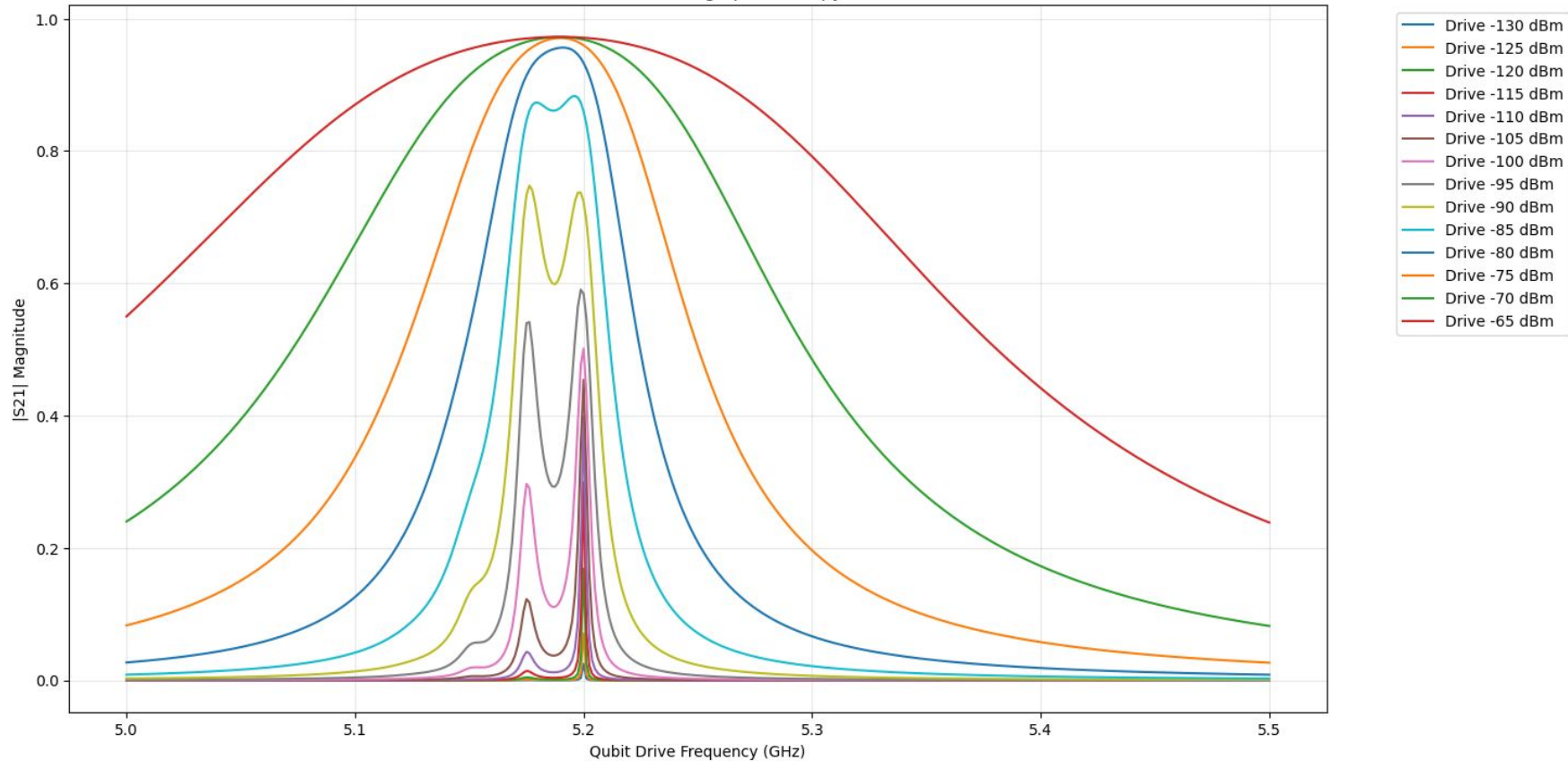


Nonlinear response emerges at high power(-80 dBm)

Observation of Power Broadening:



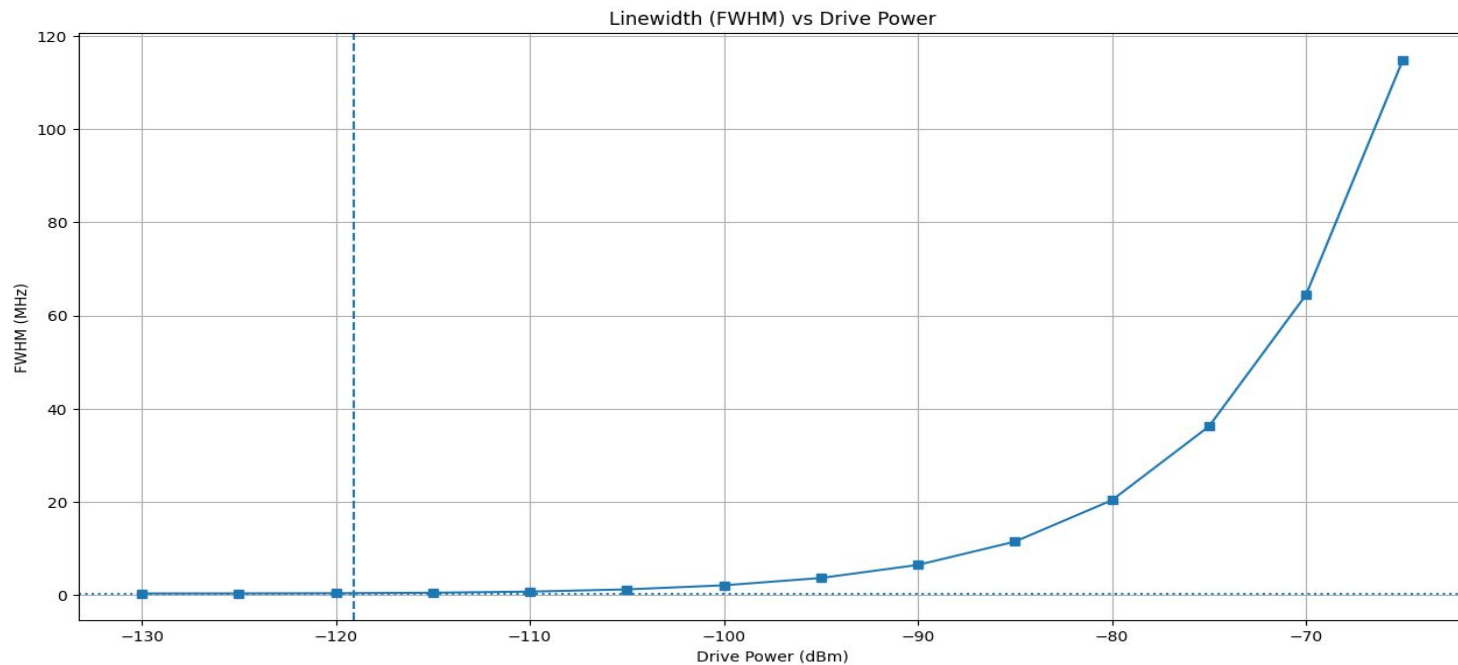
Power Broadening Spectroscopy

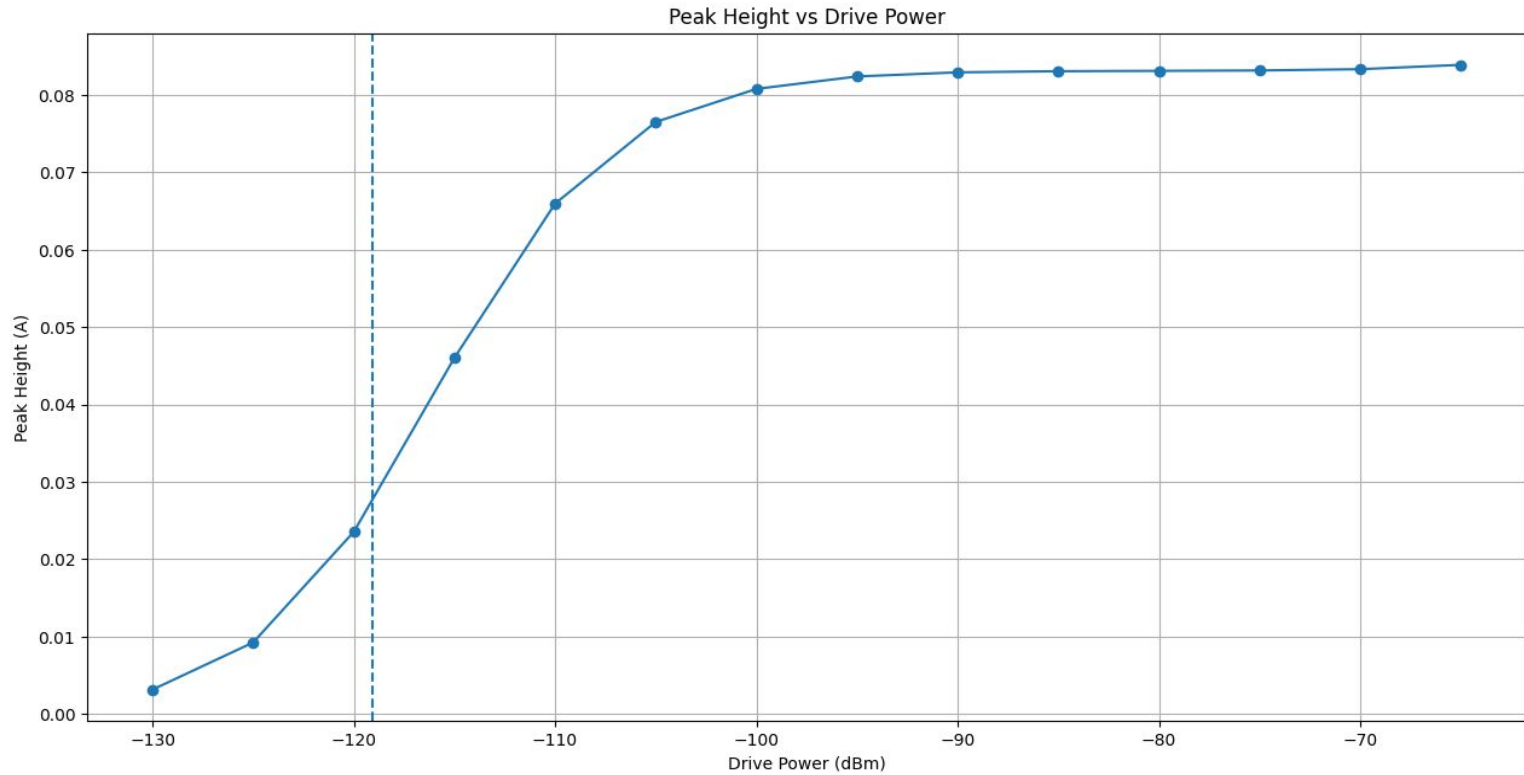


Wide frequency scan showing overall spectral response under strong drive conditions

Readout power: fixed
Qubit drive power: increased gradually

Optimal Qubit Drive Power Selection:





Optimal Power = -119dBm

Anharmonicity Analysis of the Transmon Qubit:

- A superconducting transmon is a weakly nonlinear quantum oscillator
- Realized using a Josephson junction shunted by a large capacitor
- Unlike a harmonic oscillator, it has unequally spaced energy levels
- Nonlinearity arises from the cosine-shaped Josephson potential
- This weak anharmonicity enables selective qubit control and readout
- Transmon is described by the Hamiltonian:

$$H = 4E_C(\hat{n} - n_g)^2 - E_J \cos(\hat{\phi})$$

E_C :charging energy E_J
:Josephson energy

Energy levels: $|0\rangle \rightarrow |1\rangle \rightarrow |2\rangle$
with $f_{01} \neq f_{12}$

Due to cosine nonlinearity, energy levels are not equally spaced

Transmon-Resonator System and Low-Power Spectroscopy:

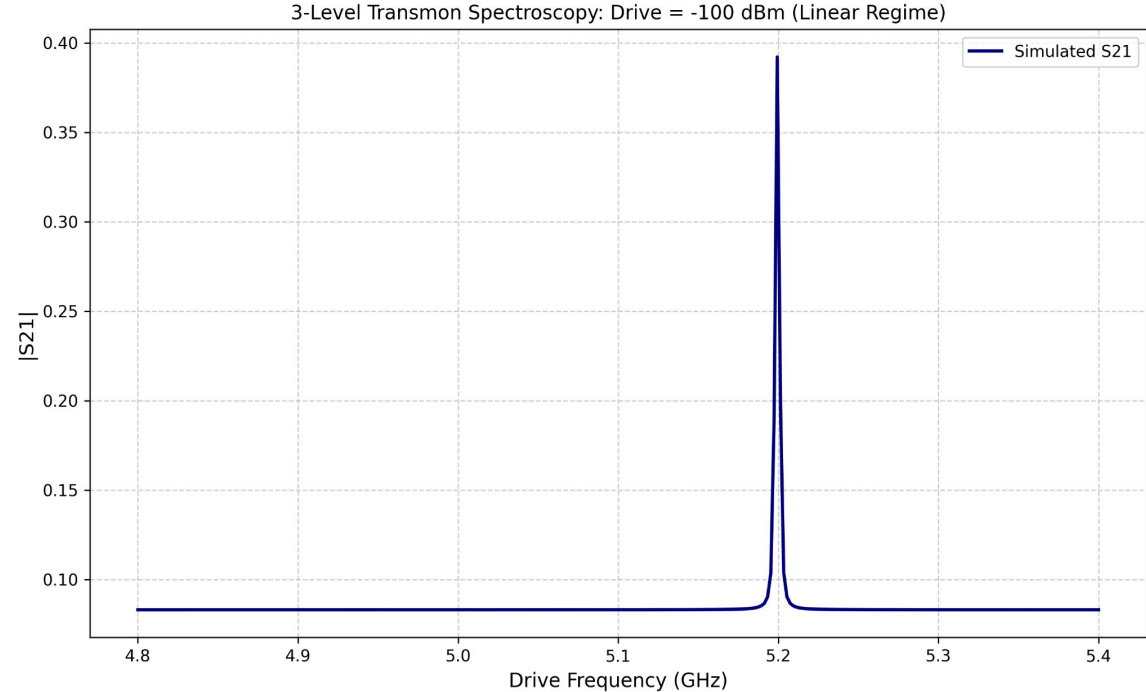
- Coupled transmon-resonator system described in circuit QED framework

- Hamiltonian:

$$H = \hbar\omega_r a^\dagger a + \hbar\omega_q b^\dagger b + \frac{\hbar\alpha}{2} b^\dagger b^\dagger b b + \hbar\chi a^\dagger a b^\dagger b$$

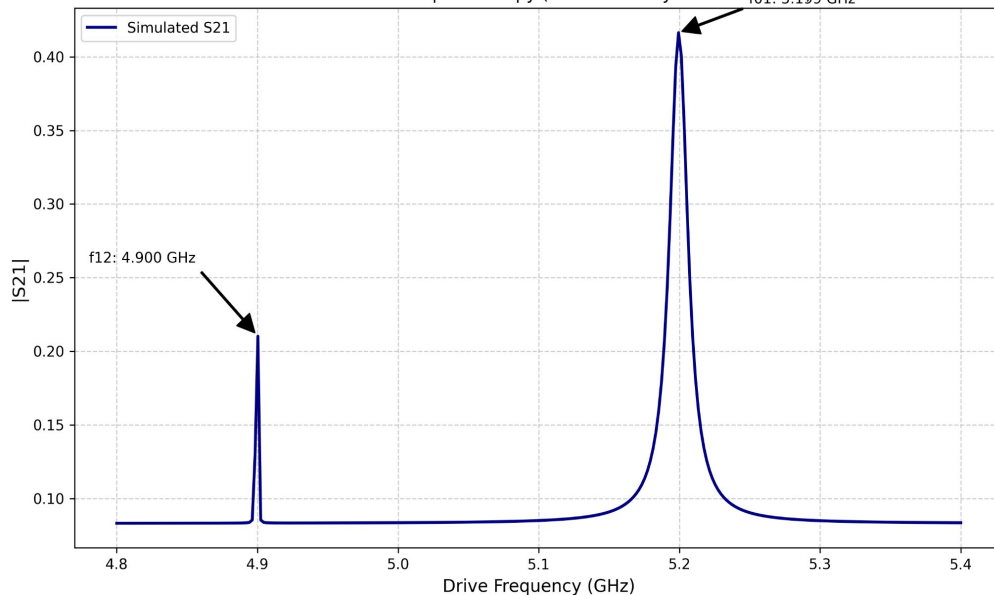
- Dispersive coupling:

$$\chi = \frac{g^2 \alpha}{\Delta(\Delta + \alpha)}$$



Low-power regime: only fundamental transition f₀₁ is visible

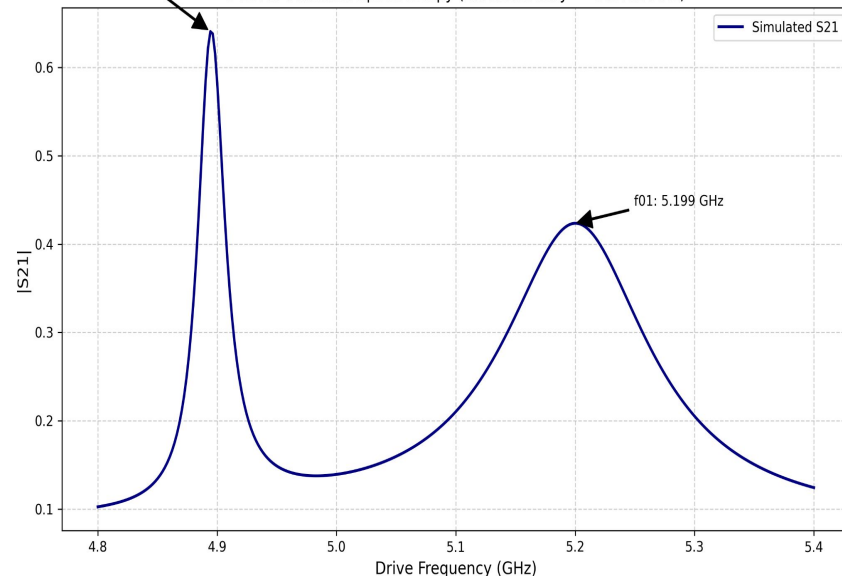
3-Level Transmon Spectroscopy (Anharmonicity = -299.00 MHz)



Emergence of f12 transition at intermediate power

Weak drive: only f01 transition is visible
 Intermediate drive: onset of f12 transition
 High drive: nonlinear broadening and spectral shift

3-Level Transmon Spectroscopy (Anharmonicity = -305.02 MHz)

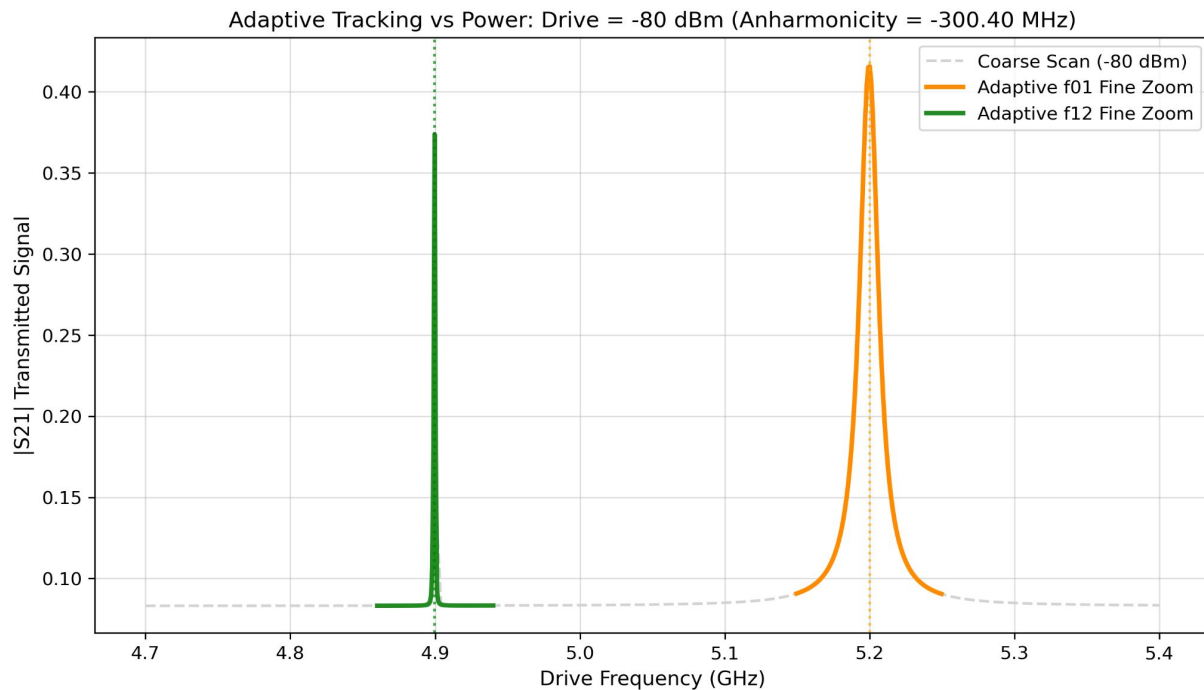


clear resolution of f01 and f12

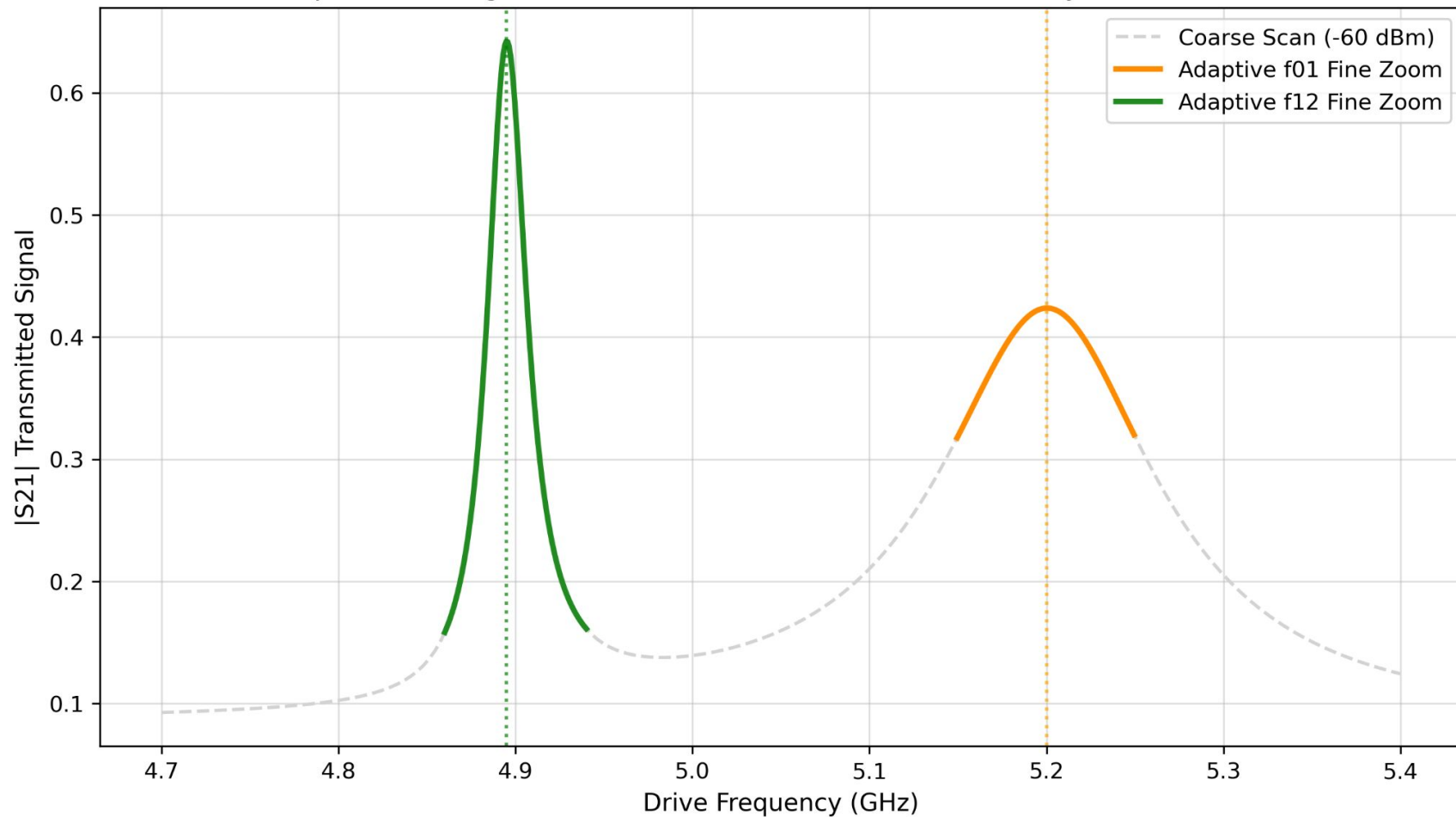
Extracted anharmonicity remains approximately consistent across drive powers

Adaptive Spectroscopy Algorithm for Efficient Frequency Estimation:

- Adaptive spectroscopy uses a coarse-to-fine frequency search strategy
- Initial coarse scan identifies approximate f01 transition
- Refined local scan improves resolution around resonance
- Second transition estimated using:
$$f_{12} \approx f_{01} + \alpha$$
- Reduces unnecessary frequency sampling



Adaptive Tracking vs Power: Drive = -60 dBm (Anharmonicity = -305.25 MHz)



Conclusion:

- Frequency-domain characterization of superconducting qubit systems was successfully performed
- Key spectroscopic phenomena observed:
 - Two-tone spectroscopy response
 - Photon-number splitting
 - Power broadening
- Optimal readout and drive powers were identified for stable measurements
- Transmon anharmonicity was extracted:
 $\alpha \approx -300$ MHz
- Adaptive method achieved same accuracy as brute-force method with improved efficiency

Future work:

- Experimental validation of spectroscopy protocols on real superconducting hardware
- Inclusion of realistic noise and decoherence models in simulations
- Development of improved spectroscopy and optimization techniques
- Exploration of quantum error mitigation strategies in superconducting architectures.

Thank You